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c) 2 orders n^2

index: 1, 2, 4, 6, ..., n

$\Theta(n \log n)$ if it runs for every n but $\Theta(n^2)$ is bigger

$\Theta(n^2)$ is bigger

d) Outer n

size: 10, 15, 22, ...

No powers, it's still $\Theta(n)$ $\Theta(n)$

Problem 4)

1, 2, 3, 4

||

5, 6

||

2, 3, 4

||

6

||

(3, 4)

5, 6

2, 3, 4

6

3, 4

Null

1, 5, 2, 6, 3, 4

c) in 2 which is 2

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1) 3

2) USC campus says emergency

3) False, it can't guarantee another exam time

4) I totally agree. I will try to do homeworks without AI and use it only to help learning.

5) False, not valid

6) up to next exam 26% penalty

Problem 2)

a) Error, repo inside repo is bad

b) Don't, leads to errors

c) Is fine, csci104-repos is a parent

d) git add

e) git log -3

f) git clone git@

Problem 3)

a) $i = i^*$ is constant time
while goes for

$$i = 2^i \geq n$$

$$i \geq \log_2 \log_2(n)$$

$$\Theta(\log \log(n))$$

b) Outer loop n times

Inner loop runs if

 $i \% (\sqrt{i}) \neq 0$ Inner loop is i^3

n

$$i = \sqrt{n}, 2\sqrt{n}, 3\sqrt{n}, \dots, n$$

So around $\frac{n}{\sqrt{n}}$

$$\text{let } j = 1, \dots, \sqrt{n}$$

$$\sum_{j=1}^{\sqrt{n}} (j\sqrt{n})^3 \approx n^{3/2} \cdot n^2$$

j=1

$$\Theta(n) + \Theta(n^{7/2}) = \Theta(n^{7/2})$$